B25ET10AS DSA, Practical - 4.

FAQ's

1) Define data, data item and entity.

→ • Data are simply values or sets of values.
• Data item is single unit of values.
• Entity is something that has certain attributes which may be assigned values.

eg

Attribute	Name	Age	City
Value	xyz	16	Pune.

2) Define data structure.

→ A way of organizing data that considers not only the items stored, but also their relationship to each other. Advance knowledge about the relationship bet" data items allows designing of efficient algorithms for the manipulation of data.

3) What are types of data structure? Give examples.

→ Linear and non-linear, primitive and non-primitive.
Linear : arrays, linked list.
non-linear : trees, graphs.

4) What is nested structure?

→ A structure within a structure in C is called a nested structure.

5) What are unions?

→ Same as structures, are constructed/user defined data type which can group together different data types.

6) Compare structure with Unions.

→ All members of a structure can be accessed at any time. Only one member of Union can be accessed at a time in case of union & other members contain garbage value.

7) What are the operators used to access data from structures?

→ The operators used to access data from the structures are dot(.) & arrow(→).

8) What is array of structure?

→ To declare an array of structures, we first define a structure and then declare an array variable of that type.

9) List out applications of structures.

→ Used in database management & forming an array of structures.

10) How to access structure member through pointer?

→ It has two ways:

i) referencing pointer to another address to access memory.

ii) Using dynamic memory allocation.